
RESEARCH PAPER

Society Under Monetary Stress VII: Borrowed Resilience

Consumer Credit, Debt Service, and Household Balance-Sheet Fragility

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Abstract

This paper describes how U.S. household credit can function as a private shock absorber without treating all debt as distress. Using public FRED series, it separates debt-service flows, credit stocks, and loan-performance indicators. The revised evidence is deliberately cautious. Full-year financial-obligation comparisons end in 2022 because the FRED F0DSP series is discontinued and has only three observations in 2023. Continuing series show that debt-service burdens in 2024 remained below pre-GFC levels, while saving rates, nonrevolving credit, student and auto loans, and rising credit-card delinquency kept household resilience dependent on serviceability, liquidity buffers, and repayment performance.

JEL codes: D14, E21, E31, E44, G51 **Keywords:** household finance, consumer credit, debt service, saving, delinquency, balance-sheet fragility

1. Introduction

Household credit is often discussed in two unsatisfying ways. One view treats debt growth as automatic evidence of household distress. Another view treats low aggregate debt-service ratios as proof that household balance sheets are healthy. Both readings are too simple. Credit can finance education, vehicles, durable goods, homes, and temporary liquidity needs. It can also allow households to defer stress when income, saving, labor margins, or public support are insufficient.

This paper uses the phrase **borrowed resilience** for that middle ground. Borrowed resilience means that private credit can absorb pressure for a period while creating future repayment exposure. It is not a claim that all borrowing is forced, harmful, or unsustainable. It is a framework for reading household balance sheets across three distinct objects: credit stocks, debt-service flows, and loan-performance indicators.

The paper is intentionally descriptive. It does not claim to identify the household-level causal effect of inflation, rent, medical bills, employment shocks, or rate changes on borrowing. Aggregate data cannot prove why a household borrowed, whether the borrowing was voluntary investment or emergency bridge financing, or which households are most exposed. The contribution is narrower: it builds a disciplined public-data map that prevents common overstatements and clarifies where a stronger microdata design would need to focus.

That discipline changes the interpretation. Debt-service and financial-obligation ratios were much higher before the global financial crisis than in the latest comparable full-year flow data. A simple claim that aggregate household debt stress is worse now than in 2007 is not supported. At the same time, the household credit system still matters. Consumer credit remains large relative to disposable income, nonrevolving credit became a larger part of the credit stack, student loans rose

substantially through 2019, auto loans remained persistent, and the saving buffer weakened after 2019. Delinquency indicators also moved up after their 2021-2022 lows, especially for credit cards, even though they remained below GFC-era stress levels in the annual data used here.

The paper therefore frames household credit as a **serviceability problem**, not a simple stock problem. A household can carry a large debt stock if rates, maturities, income, and employment remain favorable. The same balance sheet can become fragile if refinancing costs rise, saving falls, employment weakens, or delinquency transitions deteriorate. Aggregate data cannot tell the whole household story, but they can show which dimensions should not be confused.

This framing also answers the strongest objection to the paper: credit is often resilience, not fragility. That objection is correct. A credit card paid in full for timing or rewards is not the same object as a revolving balance carried because cash flow failed. A vehicle loan that preserves access to work is not the same object as high-cost emergency borrowing. A student loan can be a long-horizon investment, a repayment burden, or both. The paper therefore uses borrowed resilience as a measurement warning, not an anti-credit label.

2. Data and measurement

The analysis uses public FRED CSV downloads. The central series are household debt service payments as a percent of disposable personal income (TDSP), household financial obligations as a percent of disposable personal income (FODSP), consumer credit (TOTALSL), revolving credit (REVOLSL), nonrevolving credit (NONREVSL), student loans (SLOAS), motor vehicle loans (MVL0AS), credit-card delinquency (DRCCCLACBS), consumer-loan delinquency (DRCLACBS), mortgage delinquency (DRSFRMACBS), personal saving rate (PSAVERT), disposable personal income (DSPI), CPI (CPIAUCSL), unemployment (UNRATE), underemployment (U6RATE), the effective federal funds rate (FEDFUNDS), and the 10-year Treasury yield (DGS10).

An external-style review correctly flagged that the earlier version handled the data window too loosely. The revised version separates the **full-year financial-obligation window** from the **continuing-series update window**. In the pulled FRED data, TDSP begins in 2005. FODSP is available historically but is discontinued and ends at 2023:Q3. Therefore, full-year comparisons that require both TDSP and FODSP should end in 2022. The 2023 FODSP value is a three-observation average and is explicitly labeled partial. The 2024 values shown for continuing series exclude FODSP rather than imputing it.

This matters because a May 2026 paper should not pretend that discontinued and continuing series have the same coverage. TDSP, consumer credit, saving, income, delinquency, and rate series continue beyond 2023 in the pulled data. Student-loan and motor-vehicle-loan series used here continue through 2024:Q4 in the pulled data. The paper uses 2024 for continuing credit-stock and performance context but treats 2022 as the last clean full-year FODSP benchmark.

Credit-stock series are reported in millions of dollars. DSPi is reported in billions of dollars. The paper divides credit stocks by 1,000 before normalizing by disposable personal income. Annual values are simple within-year averages. For daily Treasury yields, annual values are averages over observed trading days.

The paper reports a debt-service-minus-saving comparison, but it is demoted from a headline buffer measure. The personal saving rate is an aggregate flow, not a liquid-asset buffer. It is strongly affected by the income distribution and by temporary fiscal and consumption movements. Subtracting it from a debt-service ratio does not measure household solvency, liquidity adequacy, emergency savings, or default risk. It is used only as a descriptive warning that payment burdens and current saving flows can move in different directions.

3. Results

Table 1 reports debt-service flows, broader financial obligations, the saving rate, and the federal funds rate. It now labels the 2023 FODSP value as partial and shows 2024 only for continuing series.

Table 1. Debt-service flows, obligations, and saving

FODSP is discontinued and has only three observations in 2023; 2024 has no FODSP value in the pulled public series.

Year	TDSP	FODSP	FODSP obs.	Saving rate	TDSP - saving	Fed funds	Note
2005	14.9%	17.5%	4	2.2%	12.7%	3.2%	full-year core
2007	15.7%	18.0%	4	2.5%	13.2%	5.0%	full-year core
2019	11.6%	14.7%	4	7.3%	4.3%	2.2%	full-year core
2022	10.6%	14.4%	4	3.4%	7.3%	1.7%	full-year core
2023	10.7%	14.1%	3	5.6%	5.2%	5.0%	partial FODSP
2024	11.1%			5.4%	5.7%	5.1%	continuing series only

The central flow finding is unchanged but now better qualified. Debt service was 15.7 percent of disposable personal income in 2007. It was 10.6 percent in 2022 and 11.1 percent in 2024. Financial obligations were 18.0 percent in 2007 and 14.4 percent in 2022. The partial 2023 financial-obligations value is 14.1 percent, but because it uses only three quarters it should not be treated as a standard full-year benchmark.

Figure 1 shows the same pattern. The full-year FODSP line stops before the continuing-series portion becomes the relevant comparison. The main point is not that households are free of stress. It is that required payment flows, measured against aggregate disposable personal income, were materially higher before the GFC than in the early 2020s.

Figure 1. Debt-service flows and the saving buffer

FODSP is discontinued; 2023 is partial and later values are omitted

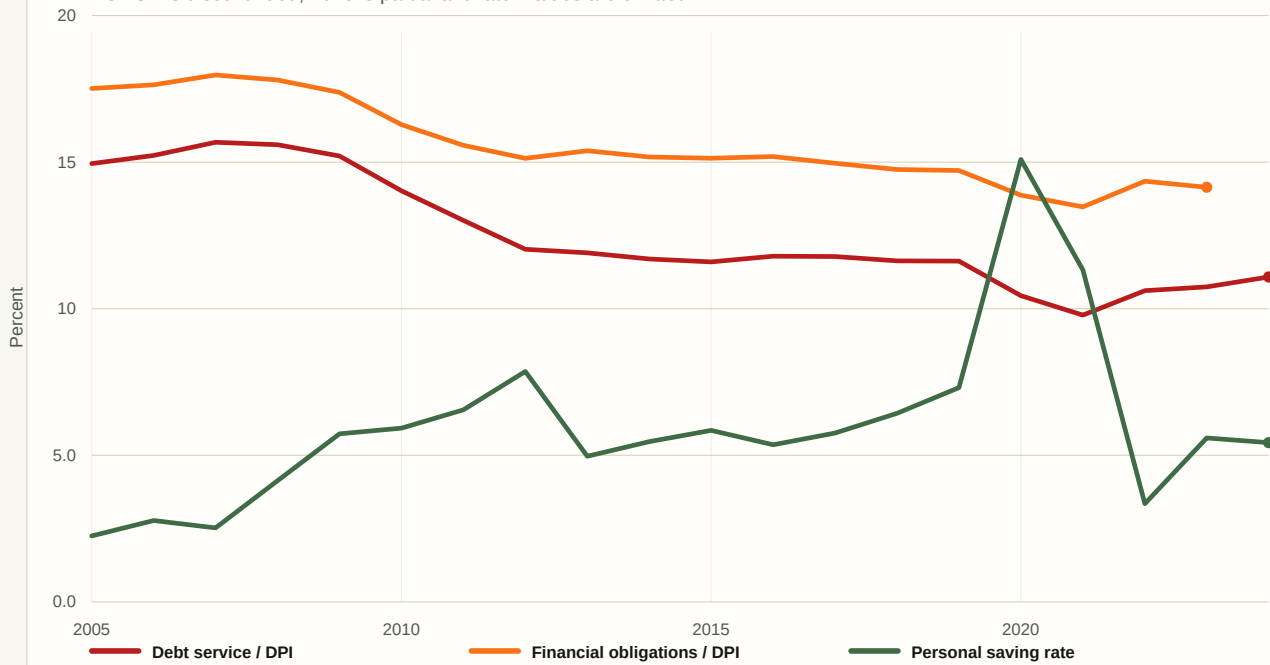


Figure 1. Debt-service flows and the saving buffer

Table 2 shifts from flows to credit composition and performance. This separation is important because consumer credit stocks do not map one-to-one onto debt-service ratios. Total consumer credit excludes mortgages, while TDSP includes mortgage and consumer debt service. Delinquency measures are bank-portfolio performance indicators, not direct household-level welfare measures.

Table 2. Credit composition and performance

Credit stocks are normalized by disposable personal income. Delinquency rates are commercial-bank portfolio measures, not household-level universal delinquency rates.

Year	Consumer credit/DPI	Revolving/DPI	Nonrevolving/DPI	Student loans/DPI	Auto loans/DPI	Credit card delinquency	Consumer-loan delinquency	Mortgage delinquency
2005	24.1%	8.7%	15.4%		8.7%	3.7%	2.8%	1.6%
2007	24.2%	9.2%	15.0%	5.4%	7.6%	4.3%	3.1%	2.5%
2019	25.4%	6.7%	18.8%	10.0%	7.2%	2.6%	2.3%	2.5%
2022	24.9%	5.9%	18.9%	9.3%	7.7%	2.0%	1.9%	1.9%
2023	23.8%	6.0%	17.8%	8.4%	7.4%	2.8%	2.4%	1.7%
2024	22.9%	6.1%	16.9%	8.0%	7.1%	3.2%	2.7%	1.7%

Total consumer credit was 24.2 percent of disposable personal income in 2007, 25.4 percent in 2019, and 22.9 percent in 2024. This does not show an aggregate consumer-credit explosion relative to disposable income. The more relevant shift is composition. Revolving credit fell from 9.2 percent of disposable personal income in 2007 to 6.1 percent in 2024. Nonrevolving credit rose from 15.0 percent to 16.9 percent over the same comparison.

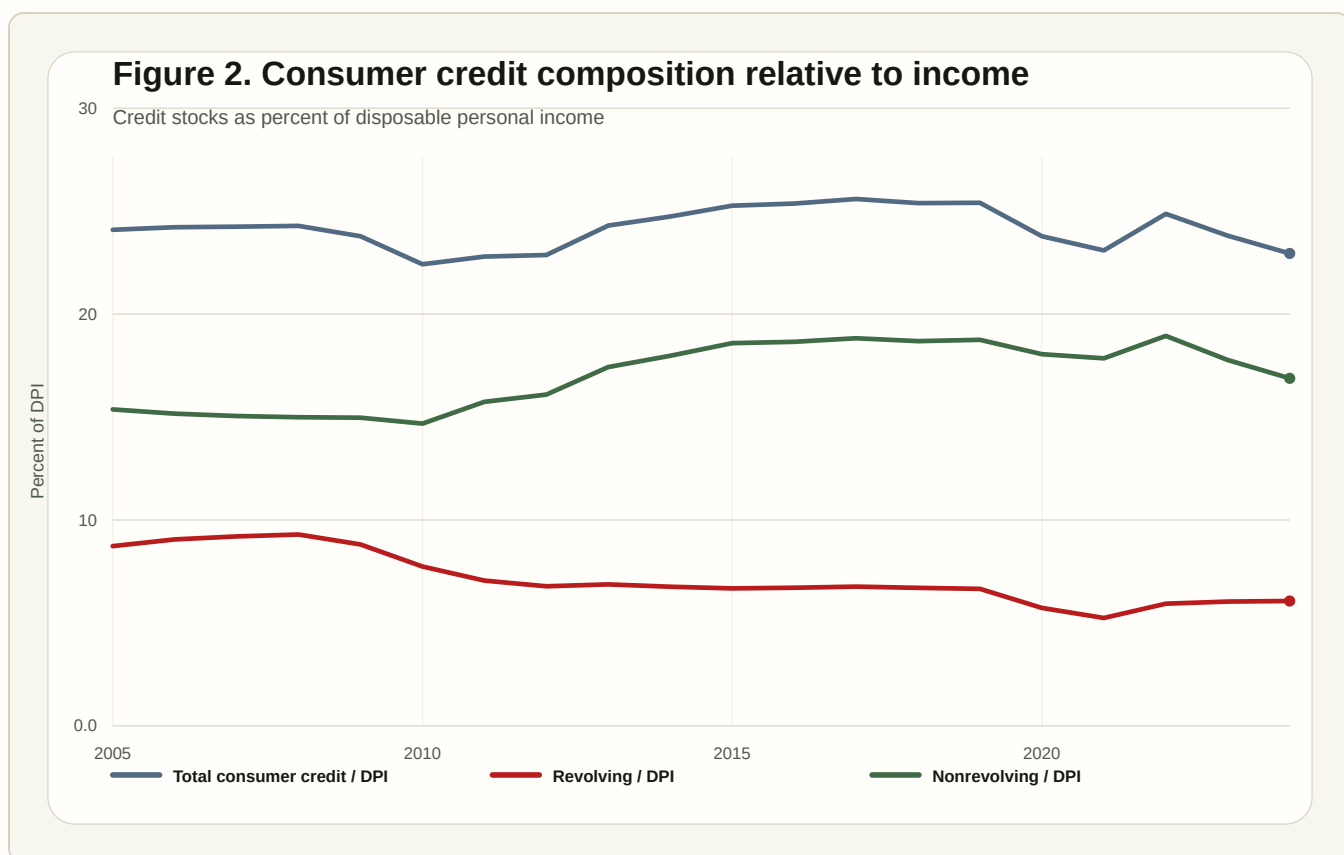


Figure 2. Consumer credit composition relative to income

Student and auto loans illustrate why composition matters. Student loans were 5.4 percent of disposable personal income in 2007 and 10.0 percent in 2019 before declining to 8.0 percent in 2024. Auto loans were 7.6 percent in 2007 and 7.1 percent in 2024. Student loans therefore mark

one of the clearest changes in the post-GFC balance-sheet mix, even though the series later eased.

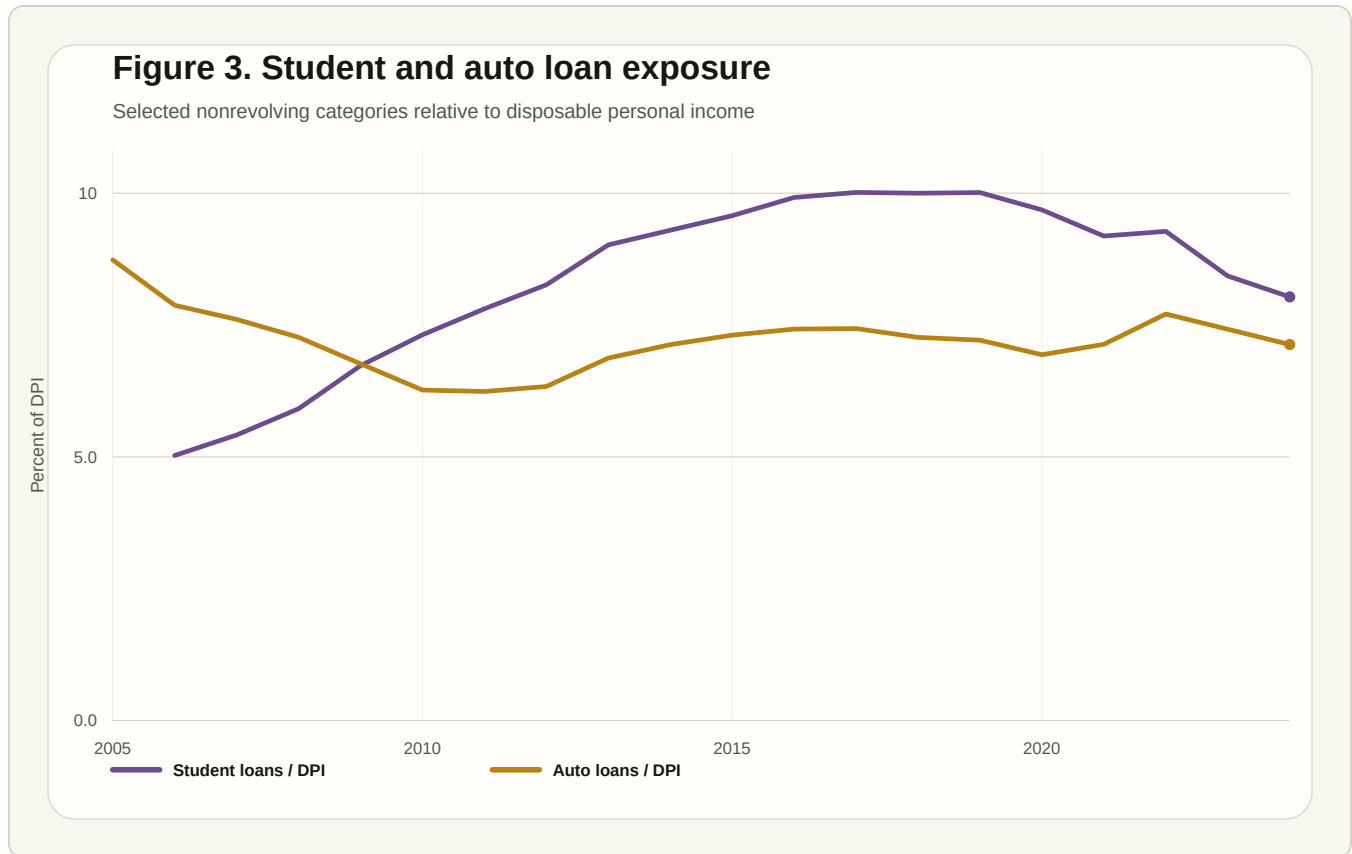


Figure 3. Student and auto loan exposure

Delinquency indicators also give a mixed picture. Credit-card delinquency was 4.3 percent in 2007, fell to 2.0 percent in 2022, and rose to 3.2 percent in 2024. Consumer-loan delinquency followed a similar but smaller move: 3.1 percent in 2007, 1.9 percent in 2022, and 2.7 percent in 2024. Mortgage delinquency was 2.5 percent in 2007 and 1.7 percent in 2024. These measures do not show GFC-level loan-performance stress, but the post-2022 rise in unsecured/consumer performance indicators deserves attention.

Figure 4. Loan-performance stress indicators

Delinquency rates on selected bank loan portfolios

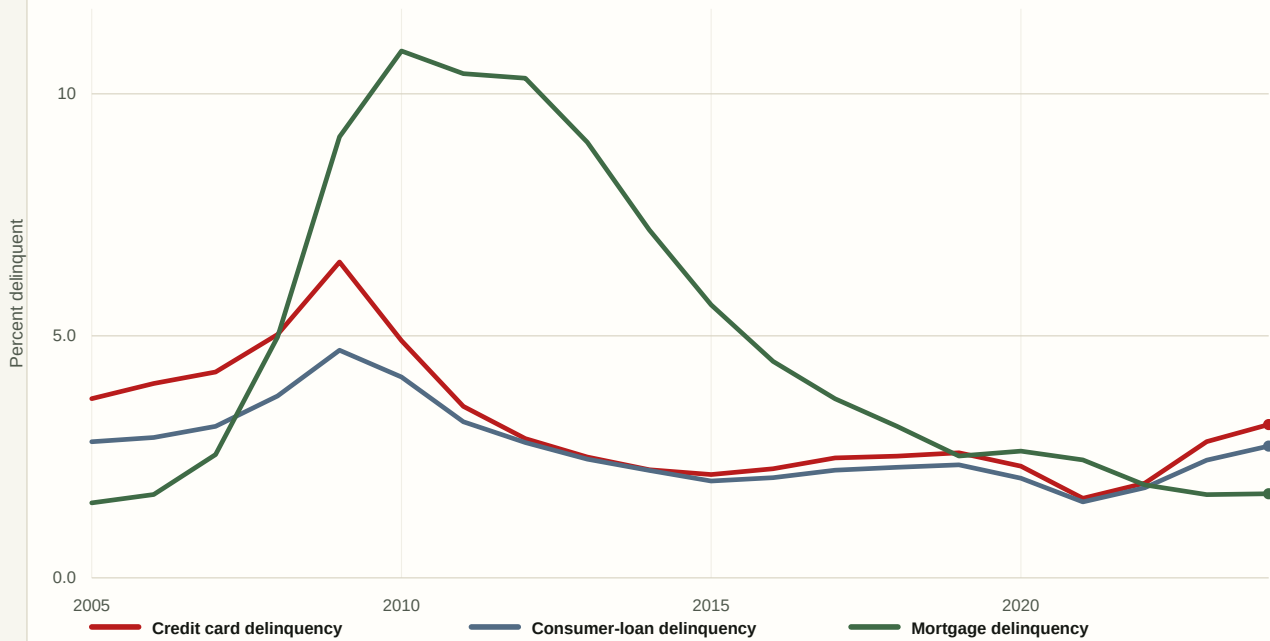


Figure 4. Loan-performance stress indicators

Figure 5 retains the obligations-versus-saving comparison only as a heuristic. The pre-GFC period combined high debt-service burden with very low saving. The 2019 period looked more comfortable. The 2022-2024 period had lower debt-service burden than 2007 but weaker saving than 2019. That is not a liquidity-buffer measure, but it helps summarize why low aggregate payment burdens do not eliminate household balance-sheet concern.

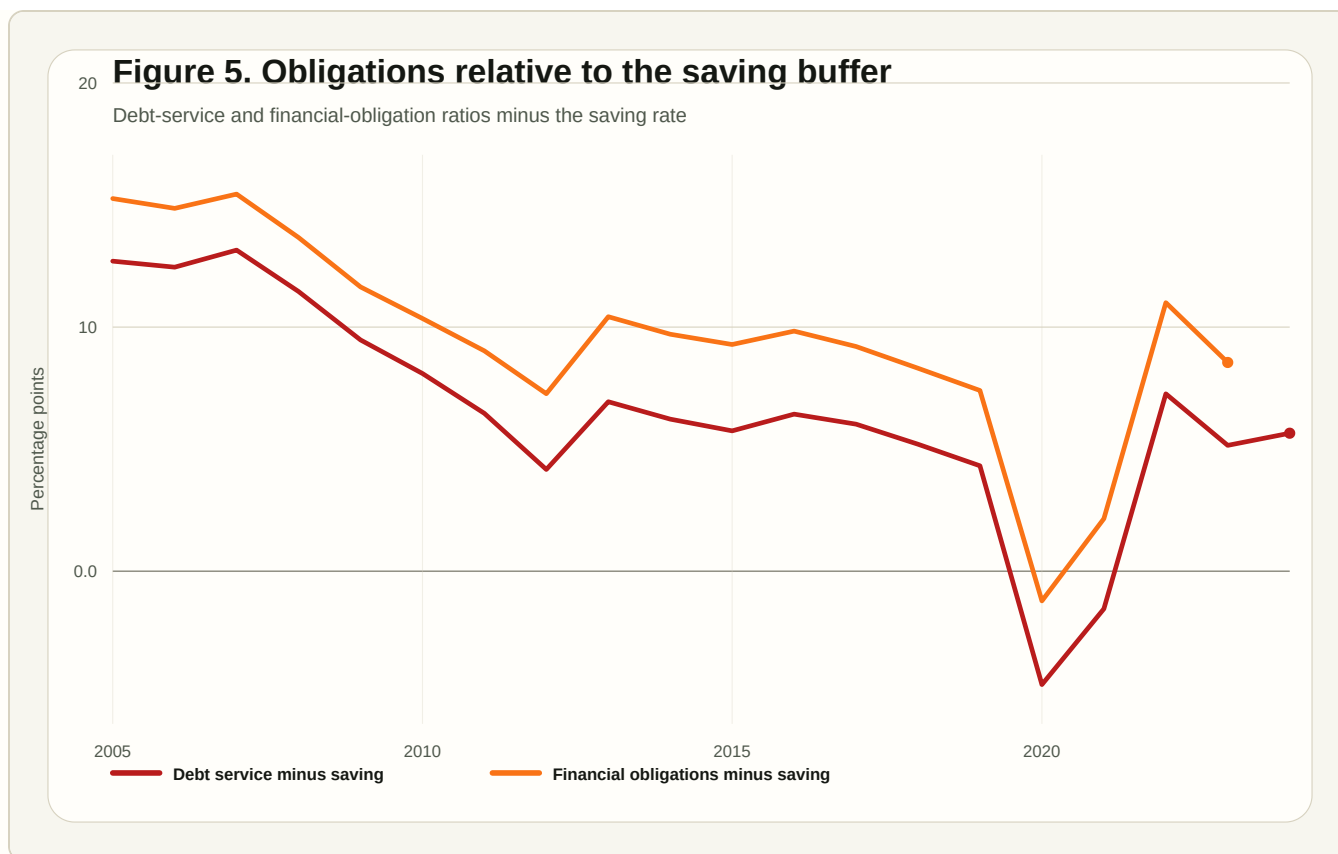


Figure 5. Obligations relative to the saving buffer

4. Interpretation: what aggregate data can and cannot show

The revised evidence supports a modest claim: aggregate household debt-service stress in the early 2020s was not a replay of 2007, but household resilience remained conditional on credit composition, saving behavior, rate conditions, and repayment performance. That claim is weaker than a causal theory of borrowed resilience, but it is more defensible.

A useful interpretation requires separating at least five credit states. **Convenience credit** is payment timing, rewards capture, or liquidity management by households that can repay in full. **Bridge credit** is short-term smoothing during a temporary gap. **Productive or durable-goods credit** finances education, vehicles, housing, repairs, or other investments that may preserve or improve future capacity. **Distress credit** is revolving, high-cost, repeated, or arrears-adjacent borrowing used because cash flow and liquid buffers are insufficient. **Credit exclusion** is the condition of having thin liquidity but no usable credit channel. These states have different welfare meanings even when they appear as the same aggregate credit stock.

The FRED series used here cannot allocate households into those states. They show aggregate stocks, payment flows, and portfolio performance. They do not show whether a balance was paid in full, carried strategically, revolved under stress, used for a productive purchase, or unavailable to

the households that needed liquidity most. That is why the paper treats borrowed resilience as a serviceability-dependence concept: the household is more exposed when normal life depends on future income, refinancing conditions, credit terms, and continued repayment capacity.

The phrase borrowed resilience should therefore be read as a conceptual frame, not an identified estimate. The data do not observe household shocks. They do not show which households borrowed after a rent increase, medical bill, employment loss, or inflation shock. They do not reveal whether borrowing stabilized consumption or raised later default risk. They do not separate households with large liquid buffers from households carrying balances because cash flow failed.

The aggregate data also hide distribution. Disposable personal income is not evenly distributed. Saving is not evenly distributed. Debt burdens are not evenly distributed. A low aggregate debt-service ratio can coexist with severe distress among renters, younger households, low-credit-score borrowers, households without liquid assets, or families facing local cost shocks. This is especially important because the households most dependent on credit smoothing may be least visible in aggregate averages.

The stock-flow-performance distinction still has value. Credit stocks show exposure. Debt-service flows show current payment burden. Delinquencies show repayment failure in measured portfolios. A complete household-finance analysis needs all three. But the aggregates are a map, not a verdict.

5. Toward a stronger empirical design

A stronger research paper would define borrowed resilience as a household-level response. For example: after an income, rent, employment, medical, or price shock, households with low liquid buffers increase credit use, reduce saving, or delay repayment more than otherwise similar households; this may stabilize consumption initially but raise later delinquency risk. That design requires microdata.

The Survey of Consumer Finances could document liquid-asset buffers, balance-sheet composition, income, and debt distribution across household types. New York Fed Consumer Credit Panel summaries or credit-bureau data could measure balances, utilization, delinquencies, and repayment transitions. The Consumer Expenditure Survey or PSID could help connect borrowing and consumption smoothing. Local rent, unemployment, or price shocks could provide variation for testing whether credit mediates household stress.

This public-data paper should therefore be read as a cleaned descriptive foundation. It identifies the right measurement distinctions, corrects the timing of discontinued series, and shows why a microdata extension is necessary. It is not a substitute for that extension.

The SHED emergency-financing work in this research program is a natural next step because it begins separating paid-in-full bridge credit from revolving credit, other emergency borrowing, inability to pay, and low-buffer/no-listed-financing states. That household-level distinction is exactly

what the aggregate balance-sheet data cannot provide.

6. Conclusion

Household credit is neither automatically harmful nor automatically benign. It is a private balance-sheet tool whose meaning depends on why borrowing occurs, who borrows, at what terms, whether repayment is feasible, and what alternatives were available. Aggregate public data cannot answer those household-level questions, but they can discipline the conversation.

The disciplined reading is mixed. Debt-service and financial-obligation burdens were much higher before the GFC than in the latest clean full-year flow comparisons. Consumer-credit stocks did not explode relative to disposable personal income. At the same time, nonrevolving credit became more important, student-loan exposure rose sharply through 2019, auto debt remained persistent, saving weakened after 2019, and credit-card delinquency rose from its 2022 low by 2024.

Borrowed resilience is therefore not a crisis claim, and it is not a claim that credit is bad. It is a warning about dependence on serviceability. When household budgets are pressured, credit can preserve normal life for a while. Whether that is convenience, healthy smoothing, productive investment, fragile deferral, or exclusion from liquidity depends on buffers, rates, income stability, credit terms, repayment performance, and household alternatives. The public aggregate data show the contours of that problem. The next research step is distributional and household-level.

References

- Dynan, Karen, Kathleen Johnson, and Karen Pence. 2003. "Recent Changes to a Measure of U.S. Household Debt Service." *Federal Reserve Bulletin* 89: 417–426.
- Federal Reserve Board. Various series. Household Debt Service and Financial Obligations Ratios; Consumer Credit; Charge-Off and Delinquency Rates. Board of Governors of the Federal Reserve System.
- Federal Reserve Bank of New York. Various years. *Quarterly Report on Household Debt and Credit*. Center for Microeconomic Data.
- Gross, David B., and Nicholas S. Souleles. 2002. "Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data." *Quarterly Journal of Economics* 117 (1): 149–185.
- Mian, Atif, and Amir Sufi. 2011. "House Prices, Home Equity-Based Borrowing, and the U.S. Household Leverage Crisis." *American Economic Review* 101 (5): 2132–2156.
- Mian, Atif, and Amir Sufi. 2014. *House of Debt*. University of Chicago Press.
- Parker, Jonathan A. 2017. "Why Don't Households Smooth Consumption? Evidence from a \$25 Million Experiment." *American Economic Journal: Macroeconomics* 9 (4): 153–183.

Sullivan, James X. 2008. "Borrowing During Unemployment: Unsecured Debt as a Safety Net." *Journal of Human Resources* 43 (2): 383–412.

Replication note

The analysis uses public FRED CSV downloads pulled in May 2026. Series include TDSP, F0DSP, TOTALSL, REVOLSL, NONREVSL, SLOAS, MVLOAS, DRCCLACBS, DRCLACBS, DRSFRMACBS, PSAVERT, DSPI, DSPIC96, MEH0INUSA646N, CPIAUCSL, UNRATE, U6RATE, FEDFUNDS, and DGS10. Monthly, quarterly, annual, and daily observations are averaged within calendar year. Credit-stock series reported in millions of dollars are divided by 1,000 before normalization by disposable personal income in billions. Full-year comparisons requiring F0DSP end in 2022. The 2023 F0DSP value is a partial-year average using Q1-Q3 only. Continuing series are carried forward where available, including 2024 credit-stock and delinquency context. The replication workflow rebuilds processed data, tables, and figures from public source series.